

World Balance Architects

A Multi-Agent Ecological Optimization and Learning
Simulation

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Project Description

This project presents a competitive AI-based ecological simulation where intelligent agents act as *planet architects* responsible for shaping and maintaining a living ecosystem. The environment consists of interconnected systems including water flow, agriculture, forests, and planetary stability parameters such as temperature and oxygen.

Agents make strategic decisions to develop resources, regulate environmental balance, and outperform opponents while preventing ecological collapse. The system models dynamic resource interactions, long-term environmental consequences, and adversarial decision-making.

The project integrates **planning-based AI**, **simulation-based AI**, and **learning-based AI**, demonstrating how different artificial intelligence paradigms perform in complex system management rather than traditional action-based gameplay.

Multi-Agent Intelligence Framework

The system compares three AI decision paradigms:

Agent	AI Method	Decision Style
Agent 1	Minimax + Alpha-Beta	Adversarial planning
Agent 2	Monte Carlo Simulation	Simulation-based planning
Agent 3	Q-Learning	Experience-based learning

State Representation

Agents observe abstracted environmental states:

- Water level category
- Food level category

- Oxygen level category
- Temperature category

State-Space Reasoning

Agents evaluate future world states generated by possible actions, similar to state-space trees used in classical AI search.

Learning Component (Q-Learning Agent)

The learning agent updates a Q-table:

State $\rightarrow$ Action $\rightarrow$ Value

based on rewards derived from environmental stability changes.

Agent Action Space

Each agent has **seven possible actions**:

1. Plant Forest
2. Build Canal
3. Build Reservoir
4. Plant Farm
5. Harvest Crop
6. Adjust Resource Allocation (Farm / Forest / Balanced)
7. Build Solar Plant

Note on State-Space Tree Diagrams

A full state-space tree with all seven actions becomes visually complex. Therefore, to clearly illustrate decision-making, the following diagrams show **2–3 representative actions**, while the agents internally consider the full action set.

Agent 1: Minimax + Alpha-Beta State Space

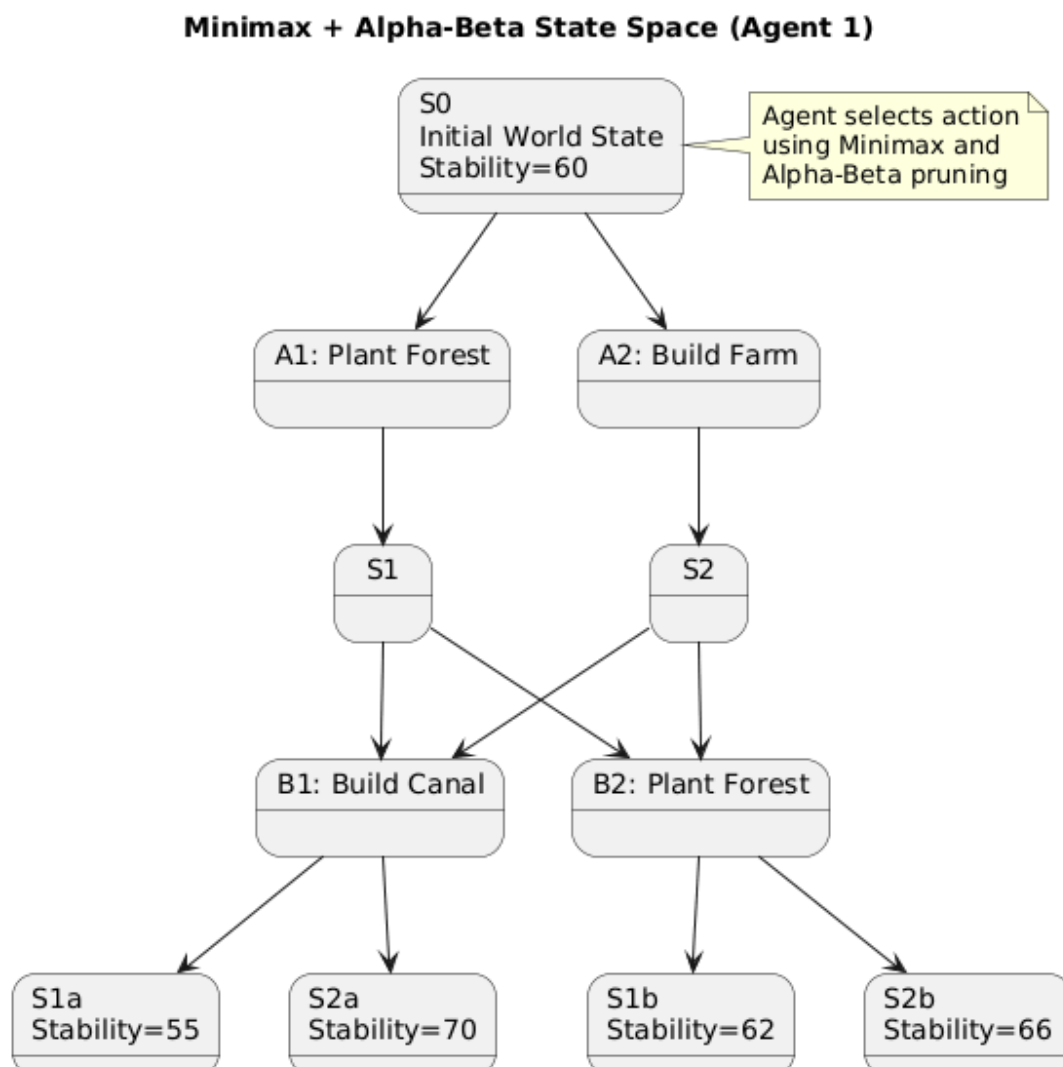


Figure 1: State-space tree showing adversarial search for the Minimax + Alpha-Beta agent.

Agent 2: Monte Carlo Simulation State Space

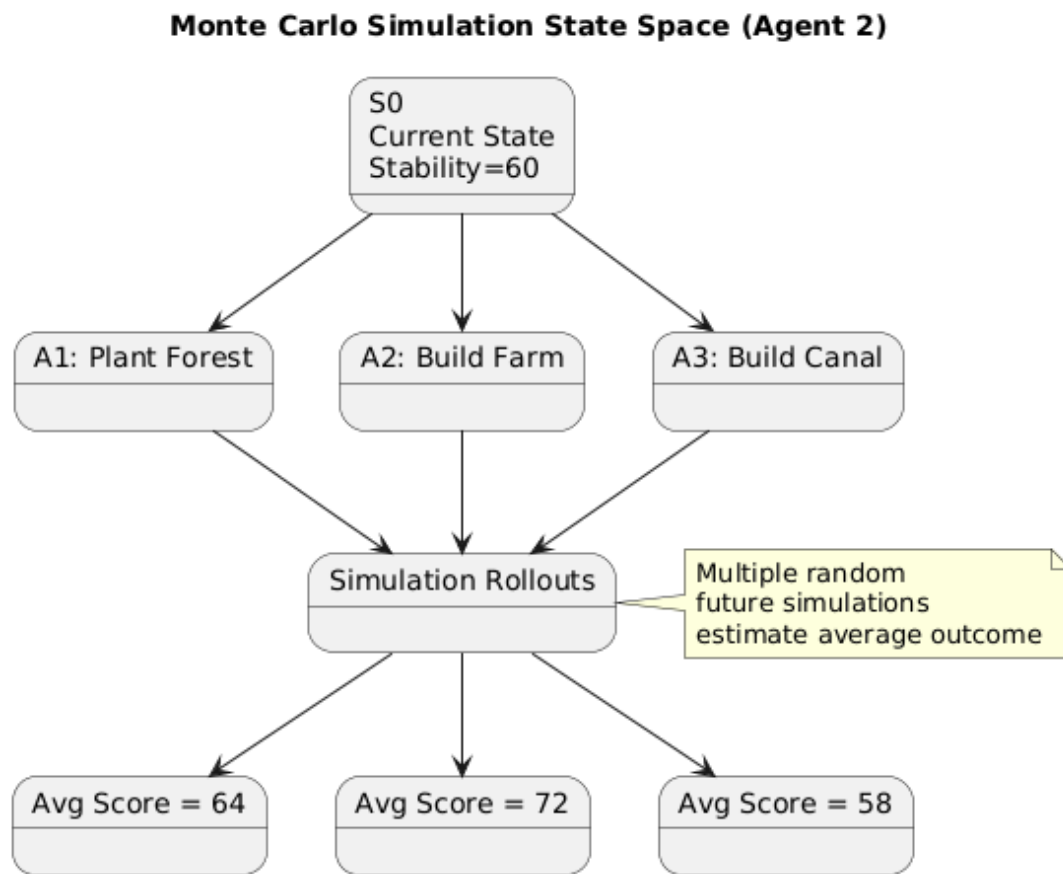


Figure 2: State-space tree illustrating simulation rollouts for the Monte Carlo agent.

Agent 3: Q-Learning State Space

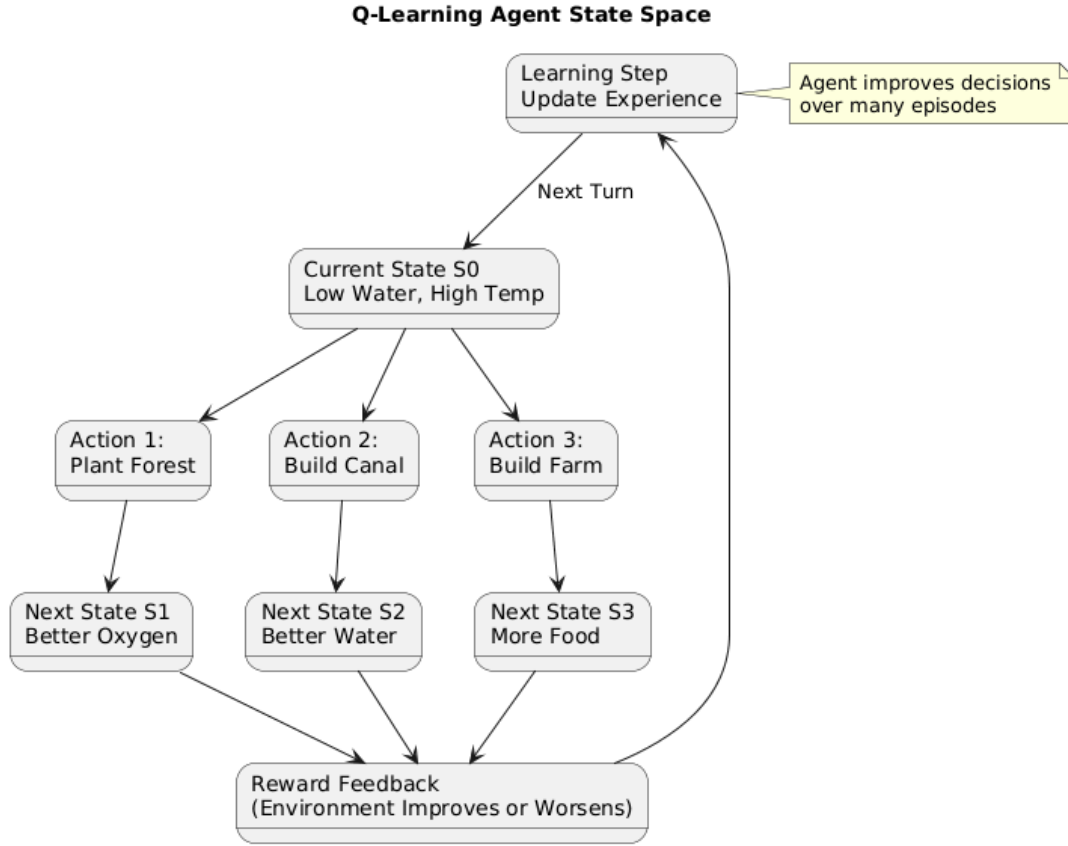


Figure 3: State-space representation of the learning cycle for the Q-Learning agent.

Game Termination and Winner Determination

The simulation runs in discrete turns and represents a dynamic ecological management scenario. The game is considered complete under the following conditions:

Game Termination Conditions

- The game reaches a predefined maximum number of turns (e.g., 50–60 turns).

- Planetary stability falls below a critical threshold, indicating ecological collapse.

Winner Determination

At the end of the simulation, the performance of each agent is evaluated using an overall stability score derived from environmental parameters and resource management effectiveness.

The winner is determined based on:

- Final planetary stability level
- Total food produced
- Oxygen balance maintained
- Temperature deviation from optimal range
- Resource efficiency (water and energy utilization)

An agent that maintains higher long-term stability while efficiently managing resources is declared the winner. In cases where ecological collapse occurs, the agent responsible for better stability preservation prior to collapse is considered superior.

This evaluation framework allows comparison of planning-based, simulation-based, and learning-based AI strategies in managing complex dynamic systems.